Experiment: 5

Simulation and Throughput Analysis of LTE-Advanced Carrier Aggregation

MATLAB Program

```
clc; clear; close all;
BW1 = 20; % Component carrier bandwidth (MHz)
SE = 4; % Spectral efficiency (bits/s/Hz)
Ncc = 1; % Number of component carriers
BWagg = BW1*Ncc;
Throughput = BWagg*SE; % Mbps for MHz × bits/s/Hz
disp('Single Component Carrier')
disp(['Bandwidth = ',num2str(BWagg),' MHz'])
disp(['Estimated Throughput = ',num2str(Throughput),' Mbps'])
figure
bar(Ncc,Throughput)
grid on
xlabel('Number of Component Carriers')
ylabel('Estimated Throughput (Mbps)')
title('LTE-Advanced Single Carrier Throughput')
```

MATLAB Program: objective 3

```
clc; clear; close all;
BW = [5 10 15 20]; % Carrier bandwidths (MHz)
SE = 4; % bits/s/Hz
TP = BW .* SE; % Estimated throughput (Mbps)
disp('Bandwidth (MHz) Throughput (Mbps)')
disp([BW' TP'])

figure
plot(BW,TP,'o-','LineWidth',2)
grid on
```

xlabel('Carrier Bandwidth (MHz)')

ylabel('Estimated Throughput (Mbps)')

title('Throughput vs Component Carrier Bandwidth')

Expected Output: Throughput increases as the component-carrier bandwidth increases.



